22/7/26 EXP NO: 4

STUDY OF LAN CONFIGURATION

Aim

To design, set-up and configure a Local Area Network (LAN)

Equipments Required

S.No	Equipment	Quantity	Specifications
1.	Personal Computers	4	ethernet (NIC) enabled
2.	Network Switch	1	8/16/24 port, unmanaged/managed
3.	Ethernet cables (coatse/cat6)	4	straight through, RJ45 connector

Implementation in CISCO Packet Tracker

Step 1: place the devices

① open cisco packet tracker

② from the panel, select end devices & drag 4 pc's onto the workspace

③ select switches and drag 1 switch

step 2: cable the devices

④ click connections → copper straight-through

⑤ click each PC, then click the switche and select an available fast ethernet

⑥ repeat for all 4 PCs

step 3: assign IP addresses to PCs

⑦ click on PC → desktop → IP config

⑧ enter static IP address

step 4: configure the switch managment IP

⑨ click the switch → CLI tab

⑩ enter the falling commands to assign a maneging IP:

```
enable
config terminal
interface WAN
IP address 10.1.1.5
no shut down
exit
```

step 5: verify connectivity

⑪ click PC → desktop → command prompt

⑫ use ping command → test connectivity.

⑬ successful "reply from .."

Step 6: demonstrated data sharing

(14) add server and connect to switch.

(15) assign it an IP address ⇒ IP config

(16) open server's services tab & enable HTTP

(17) from another ~~device~~ PC, open web and enter the server's IP

Step 7: observe packet flow:

(18) switch to simulation mode

(19) send a ping or web request.

IP Address configuration

device	IP Address	Subnet mask	connected to
~~PC~~ PC0	10.1.1.1	255.0.0.0	switch port Fa 0/1
PC1	10.1.1.2	255.0.0.0	switch port Fa 0/2
PC2	10.1.1.3	255.0.0.0	switch port Fa 0/3
PC3	10.1.1.4	255.0.0.0	~~VLAN1 interface~~ switch port Fa 0/4
switch (managment)	10.1.1.5	255.0.0.0	VLAN 1 interface

Result

Thus the study of set up and config of LAN